

Exercise 4 - Joins Exercise

Created Table- Users

```
describe users;
select * from users;
```

-- PART A - INNER JOIN

Add parameter

Table ▾ +

	¹ ₃ user_id	^A _C user_name	^A _C country
1	1	Nomvula	Johannesburg
2	2	David	Cape Town
3	3	Anele	Durban
4	4	Kabelo	Pretoria
5	5	Lerato	Port Elizabeth

Question 1

-- PART A - INNER JOIN

-- Q1: Every user who has a subscription.

```
SELECT u.user_id, u.user_name, s.subscription_id, s.start_date
FROM users u
INNER JOIN subscriptions s ON u.user_id = s.user_id;
```

Add parameter

← Results 11 of 30 →

Table ▾ +

	¹ ₃ user_id	^A _C user_name	¹ ₃ subscription_id	 start_date
1	1	Nomvula	503	2026-03-10
2	2	David	502	2026-02-01
3	3	Anele	505	2026-04-05
4	1	Nomvula	501	2026-01-15

Question 2

```
-- Q2: Every subscription with its matching plan name and price.
SELECT s.subscription_id, s.user_id, p.plan_name, p.monthly_price
FROM subscriptions s
INNER JOIN plans p ON s.plan_id = p.plan_id;
```

Add parameter

← Results 12 of 30 →

Table ▾ +

	¹ ₃ subscription_id	¹ ₃ user_id	^A _C plan_name	¹ ₃ monthly_price
1	501	1	Basic	79
2	504	6	Standard	129
3	503	1	Premium	199
4	505	3	Family	249
5	502	2	Standard	129

Question 3

```
-- Q3: Every viewing session that has a matching show.
SELECT vs.session_id, vs.user_id, sh.show_title, sh.genre, vs.watch_minutes
FROM viewing_sessions vs
INNER JOIN shows sh ON vs.show_id = sh.show_id;
```

Add parameter

← Results 13 of 30 →

Table ▾ +

	¹ ₃ session_id	¹ ₃ user_id	^A _C show_title	^A _C genre	¹ ₃ watch_minutes
1	904	7	Comedy Hour	Comedy	20
2	903	1	Crime Time	Drama	60
3	902	2	Tech Tales	Documentary	30
4	901	1	Comedy Hour	Comedy	45

Question 4

```
-- Q4: Every viewing session with the user who watched it (matching users only).
SELECT u.user_name, u.country, vs.session_id, vs.show_id, vs.watch_minutes
FROM users u
INNER JOIN viewing_sessions vs ON u.user_id = vs.user_id;
```

Add parameter

← Results 13 of 30 →

Table ▾ +

	¹ ₃ session_id	¹ ₃ user_id	^A _C show_title	^A _C genre	¹ ₃ watch_minutes
1	904	7	Comedy Hour	Comedy	20
2	903	1	Crime Time	Drama	60
3	902	2	Tech Tales	Documentary	30
4	901	1	Comedy Hour	Comedy	45

Question 5

```
-- Q5: Users with their subscriptions, plan name, and price (matches only).
SELECT u.user_name, u.country, p.plan_name, p.monthly_price, s.start_date
FROM users u
INNER JOIN subscriptions s ON u.user_id = s.user_id
INNER JOIN plans p ON s.plan_id = p.plan_id;
```

Add parameter

← Results 14 of 30 →		Table ▾	+		
	^A _C user_name	^A _C country	¹ ₃ session_id	¹ ₃ show_id	¹ ₃ watch_minutes
1	Nomvula	Johannesburg	903	702	60
2	David	Cape Town	902	703	30
3	Anele	Durban	905	705	90
4	Nomvula	Johannesburg	901	701	45

Question 6

```
-- =====
-- PART B – LEFT JOIN
-- =====

-- Q6: Every user and any subscriptions they have.
SELECT u.user_id, u.user_name, s.subscription_id, s.start_date
FROM users u
LEFT JOIN subscriptions s ON u.user_id = s.user_id;
```

Add parameter

← Results 15 of 30 →		Table ▾	+		
	^A _C user_name	^A _C country	^A _C plan_name	¹ ₃ monthly_price	¹ ₃ start_date
1	Nomvula	Johannesburg	Premium	199	2026-03-10
2	David	Cape Town	Standard	129	2026-02-01
3	Anele	Durban	Family	249	2026-04-05
4	Nomvula	Johannesburg	Basic	79	2026-01-15

Question 7

```
-- Q7: Every plan and the subscriptions on it.
SELECT p.plan_id, p.plan_name, s.subscription_id, s.user_id
FROM plans p
LEFT JOIN subscriptions s ON p.plan_id = s.plan_id;
```

Add parameter

← Results 17 of 30 →		Table ▾	+	
	¹ ₃ plan_id	^A _C plan_name	¹ ₃ subscription_id	¹ ₃ user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	12	Premium	503	1
4	13	Family	505	3
5	14	Mobile	null	null
6	11	Standard	502	2

Question 8

```
-- Q8: Every show and any viewing sessions on it.
SELECT sh.show_id, sh.show_title, vs.session_id, vs.watch_minutes
FROM shows sh
LEFT JOIN viewing_sessions vs ON sh.show_id = vs.show_id;
```

Add parameter

← Results 18 of 30 →		Table ▾	+	
	¹ ₃ show_id	^A _C show_title	¹ ₃ session_id	¹ ₃ watch_minutes
1	701	Comedy Hour	904	20
2	702	Crime Time	903	60
3	703	Tech Tales	902	30
4	704	Cooking Lab	null	null
5	706	Wild Earth	null	null
6	701	Comedy Hour	901	45

Question 9

```
-- Q9: Every viewing session and the user who watched it (NULL if user missing).
SELECT vs.session_id, vs.show_id, vs.watch_minutes, u.user_id, u.user_name
FROM viewing_sessions vs
LEFT JOIN users u ON vs.user_id = u.user_id;
```

Add parameter

←

Results 19 of 30

→

Table

▼

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	¹ ₃ session_id	¹ ₃ show_id	¹ ₃ watch_minutes	¹ ₃ user_id	^A _C user_name
1	901	701	45	1	Nomvula
2	902	703	30	2	David
3	903	702	60	1	Nomvula
4	904	701	20	null	null
5	905	705	90	3	Anele

Question 10

-- Q10: Every user, their plan (if any), and the monthly price.
SELECT u.user_name, u.country, p.plan_name, p.monthly_price
FROM users u
LEFT JOIN subscriptions s ON u.user_id = s.user_id
LEFT JOIN plans p ON s.plan_id = p.plan_id;

Add parameter

← Results 20 of 30 →

Table

+

	A ^B _C user_name	A ^B _C country	A ^B _C plan_name	1 ² ₃ monthly_price
1	Nomvula	Johannesburg	Premium	199
2	David	Cape Town	Standard	129
3	Anele	Durban	Family	249
4	Kabelo	Pretoria	null	null
5	Lerato	Port Elizabeth	null	null
6	Nomvula	Johannesburg	Basic	79

Question 11

-- =====
-- PART C - FULL OUTER JOIN
-- =====

-- Q11: Every user and every subscription, unmatched rows on either side included.
SELECT u.user_id, u.user_name, s.subscription_id, s.start_date
FROM users u
FULL OUTER JOIN subscriptions s ON u.user_id = s.user_id;

Add parameter

← Results 21 of 30 →

Table

+

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	1 ² ₃ user_id	A ^B _C user_name	1 ² ₃ subscription_id	📅 start_date
1	5	Lerato	null	null
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	null	null
6	1	Nomvula	501	2026-01-15
7	null	null	504	2026-03-20

Question 12

```
-- Q12: Every plan and every subscription, unmatched rows on either side included.
SELECT p.plan_id, p.plan_name, s.subscription_id, s.user_id
FROM plans p
FULL OUTER JOIN subscriptions s ON p.plan_id = s.plan_id;
```

Add parameter

← Results 22 of 30 → Table ▾ +					
	¹ ₃ plan_id	^A _C plan_name	¹ ₃ subscription_id	¹ ₃ user_id	
1	12	Premium	503	1	
2	10	Basic	501	1	
3	13	Family	505	3	
4	14	Mobile	null	null	
5	11	Standard	504	6	
6	11	Standard	502	2	

Question 13

```
-- Q13: Every show and every viewing session, unmatched rows on either side included.
SELECT sh.show_id, sh.show_title, vs.session_id, vs.watch_minutes
FROM shows sh
FULL OUTER JOIN viewing_sessions vs ON sh.show_id = vs.show_id;
```

Add parameter

← Results 23 of 30 → Table ▾ +					
	¹ ₃ show_id	^A _C show_title	¹ ₃ session_id	¹ ₃ watch_minutes	
1	704	Cooking Lab	null	null	
2	703	Tech Tales	902	30	
3	701	Comedy Hour	904	20	
4	702	Crime Time	903	60	
5	706	Wild Earth	null	null	
6	701	Comedy Hour	901	45	
7	null	null	905	90	

Question 14

```
-- Q14: Every user and every viewing session, unmatched rows on either side included.
SELECT u.user_id, u.user_name, vs.session_id, vs.show_id, vs.watch_minutes
FROM users u
FULL OUTER JOIN viewing_sessions vs ON u.user_id = vs.user_id;
```

Add parameter

← Results 24 of 30 → Table ▾ +						
	¹ ₃ user_id	^A _C user_name	¹ ₃ session_id	¹ ₃ show_id	¹ ₃ watch_minutes	
1	5	Lerato	null	null	null	
2	1	Nomvula	903	702	60	
3	3	Anele	905	705	90	
4	2	David	902	703	30	
5	4	Kabelo	null	null	null	
6	1	Nomvula	901	701	45	
7	null	null	904	701	20	

Question 15

```
-- Q15: Every user, every subscription, and every plan – FULL OUTER JOIN throughout.
SELECT u.user_id, u.user_name, s.subscription_id, p.plan_id, p.plan_name
FROM users u
FULL OUTER JOIN subscriptions s ON u.user_id = s.user_id
FULL OUTER JOIN plans p ON s.plan_id = p.plan_id;
```

Add parameter

← Results 25 of 30 → Table ▾ +						
	¹ ₃ user_id	^A _C user_name	¹ ₃ subscription_id	¹ ₃ plan_id	^A _C plan_name	
1	1	Nomvula	503	12	Premium	
2	1	Nomvula	501	10	Basic	
3	3	Anele	505	13	Family	
4	5	Lerato	null	null	null	
5	4	Kabelo	null	null	null	
6	2	David	502	11	Standard	
7	null	null	504	11	Standard	

Bonus Questions

```
-- =====
-- BONUS CHALLENGE
-- =====

-- Bonus 1: Which users have not subscribed to any plan?
-- Expected: Kabelo (4), Lerato (5)
SELECT u.user_id, u.user_name
FROM users u
LEFT JOIN subscriptions s ON u.user_id = s.user_id
WHERE s.subscription_id IS NULL;
```

Add parameter

← Results 26 of 30 →		Table ▾	+
	¹ ₂ user_id	^A _C user_name	
1	4	Kabelo	
2	5	Lerato	

```
-- Bonus 2: Which subscriptions reference users that do not exist?
-- Expected: subscription 504 (user_id 6)
SELECT s.subscription_id, s.user_id
FROM subscriptions s
LEFT JOIN users u ON s.user_id = u.user_id
WHERE u.user_id IS NULL;
```

Add parameter

← Results 26 of 30 →		Table ▾	+
	¹ ₂ user_id	^A _C user_name	
1	4	Kabelo	
2	5	Lerato	

```
-- Bonus 3: Which shows have never been watched?
-- Expected: Cooking Lab (704), Wild Earth (706)
SELECT sh.show_id, sh.show_title
FROM shows sh
LEFT JOIN viewing_sessions vs ON sh.show_id = vs.show_id
WHERE vs.session_id IS NULL;
```

Add parameter

← Results 27 of 30 →		Table ▾	+
	¹ ₂ subscription_id	¹ ₂ user_id	
1	504	6	


```

-- Bonus 4: Which viewing sessions reference shows that do not exist?
-- Expected: session 905 (show_id 705)
SELECT vs.session_id, vs.show_id
FROM viewing_sessions vs
LEFT JOIN shows sh ON vs.show_id = sh.show_id
WHERE sh.show_id IS NULL;

```

Add parameter

← Results 29 of 30 → Table ▾ +

	¹ ₃ session_id	¹ ₃ show_id
1	905	705

```

-- Bonus 5: Which plans have no subscribers?
-- Expected: Mobile (14)
SELECT p.plan_id, p.plan_name
FROM plans p
LEFT JOIN subscriptions s ON p.plan_id = s.plan_id
WHERE s.subscription_id IS NULL;

```

Add parameter

← Results 30 of 30 → Table ▾ +

	¹ ₃ plan_id	^A _C plan_name
1	14	Mobile